Analysis: boon-academy-intervention

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Diagnosis

Boon Academy runs 20 campuses serving roughly 300 students. Facilitator intervention rate is currently ~30% against an 80%+ target. The gap exists because facilitators lack a fast, prioritised view of which students to contact and what to say. This pipeline closes that gap by scoring every student daily, ranking them by risk, and generating draft WhatsApp messages that facilitators send in one click.

What You Found

Student intake: 300 students processed in this run. Risk distribution is available in the intervention_priority_list.xlsx output (colour-coded by CRITICAL / HIGH / MEDIUM / LOW). All data quality issues (missing metrics, type-mismatch strings, blank notes) were auto-resolved by the ingestion layer - no manual cleanup required.

What You Built

A five-stage pipeline: ingest -> score -> LLM enrich -> outputs -> docs. LLM usage this run: 15 API calls, 32994 tokens total (20893 input + 12101 output). Fallbacks triggered: 10. Output files: intervention_priority_list.xlsx, campus_dashboard_<id>.xlsx, whatsapp_messages.csv, dashboard.html, intervention_report.pdf, docs/ (nine technical docs).

What You Cut

No ML model - deterministic weighted scoring is auditable with no training data needed. No real-time server - on-demand script run fits current workflow. No OAuth or SSO - outputs are file-based, not a web app. No Docker or Kubernetes - single-machine Python script. No direct WhatsApp API sending - facilitators copy-paste from CSV as the safe v1 path.

What Next

1. Hook up real student data - replace synthetic CSVs with live exports.
2. Validate risk weights with the academic director.
3. Confirm Arabic dialect per campus (Modern Standard vs. Gulf).
4. Test outputs on LibreOffice - facilitator PCs may not have Excel.
5. Consider a scheduled daily trigger (cron or Windows Task Scheduler).